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Overview

The Dock Motor Speed Procedure is required as part of setup and calibration. Procedures for opening magnet enclosure covers and adjusting patient dock hardware (Motor On Switch Actuation Adjustment and Dock Motor Gear Assembly Adjustments) are provided in case related dock problems arise.

1- DOCK MOTOR SPEED ADJUSTMENT

Motor speed is affected by magnetic field effects on the motor and the moving parts in the patient transport. The dock motor speed cannot be factory preset; it must be adjusted for each individual site. The motor speed is set for an unloaded condition that equates to 360 rpm (or six revolutions per second). The adjustment can be made using the preferred *Section 2-2 Two-Man Procedure* or, when short handed, *Section 2-3 One-Man Procedure*. A DVM is required to verify power is off. The MEPS measurement for the Dock Motor Speed Adjustment requires one oscilloscope probe. The SSM differential measurement for the Dock Motor Speed Adjustment requires two oscilloscope probes because the Power Supply circuitry is floating.

1-1 Two-Man Procedure



The system must be set up for a Body mode protocol and the [Prepare to Scan] softkey selected initially. Pulse Body mode once at TG of 0. Do NOT open the SSM top cover for 1 minute after the system has transitioned from Head, Surface, M/C or MNS mode to Body mode. Failure to follow these instructions has resulted in the Dynamic Disable IGBT Circuitry (Direct Drive, Body1, Body2) overheating and shorting requiring CPD replacement.

1. Set up a Body mode protocol and select **[Prepare to Scan]** softkey. Wait 5 minutes before opening the SSM cover. (See above Caution for detail.)
2. Dock the patient transport, and remove the head coil carriage from the patient table if necessary. Lower table to fully down position using the foot pedals at either end of the patient transport.
3. Remove power to dock. Refer to the Safety procedure, *Section 6 - OSHA Lock-Out/Tag-Out Requirements* (SYSMSA2.DOC).
4. Lock-Out / Tag-Out circuit breaker MR1 A14 CB6 at rear of **RF/Pen Cabinet**, or circuit breaker MR1 A20 CB4 at rear of SSM in the **RF/Pen II** and **RF/PDU Cabinet**.
5. Verify that voltage between Dock and Dock RTN is **0 VDC** using a Digital Volt Meter (DVM):

RF/Pen Cabinet: Locate the MEPS Module at the bottom of the Cabinet and the test point strip at the center of the rear panel of the MEPS module. Connect the DVM.

RF/Pen II and RF/PDU Cabinet: Locate the SSM at the bottom of the Cabinet and the test point strip at the center of the rear panel of the SSM. Connect the DVM.

6. Remove the four screws and four No. 10 flat washers securing the MEPS or SSM module to the cabinet.
7. Slide the MEPS or SSM module out the front of the cabinet.

RF/Pen Cabinet:

- a. Loosen the three 4-40 captive screws on the top cover of the MEPS module. Open and remove the lid. Locate the High Current power supply board that contains the Voltage Potentiometer R38.
- b. Pull out the RFSC and loosen the captive screws on the top cover of the SSM to gain access to the white plunger switch. Open the lid, and pull the High Voltage Enable Plunger Switch to the **Up** position.
- c. Connect Channel 1 oscilloscope lead to TP3 (Dock 38.5V), and connect Channel 1 oscilloscope ground reference lead to TP1 or TP4 or TP7 (SGND).
- d. Set Channel 1 trace to zero reference graticle on the oscilloscope display. Adjust time/div to approximately **100 mS/div**. Set oscilloscope channels to **AC Coupling**.

RF/Pen II Cabinet and RF/PDU Cabinet:

- a. Do **not** disable the HV at the front of the SSM.
- b. Remove the twelve 4-40 screws on the top cover of the SSM. Open the lid, and locate the Dock/Light power supply board that contains the Voltage Potentiometer R80.
- c. Connect Channel 1 oscilloscope lead to TP3 (Dock 38.5V), and connect Channel 2 oscilloscope lead to TP4 (DOCK-RTN).
- d. Connect both oscilloscope lead grounds together. Isolate ground leads with tape as required to ensure no contact with electrical circuitry.
- e. Set both channels to ground reference. Set Channel 2 to **add/invert**, and adjust both ground leads to the same zero reference graticle on the oscilloscope display. Adjust time/div to approximately **100 mS/div**. Set oscilloscope channels to **AC Coupling**.



SHOCK HAZARD! THE SSM IPS ASSEMBLY CIRCUITRY IS FLOATING. DO NOT REFERENCE TEST EQUIPMENT TO GROUND CHASSIS OR ANY GROUND TEST POINTS.

8. Lower Table to the full down position using the Table Down pedal.
9. Switch on circuit breaker MR1 A14 CB6 on the **RF/Pen Cabinet**, or switch on circuit breaker MR1 A20 CB4 on the **RF/Pen II and RF/PDU Cabinet**.

Note

Only perform this procedure with the system set up for a Body mode protocol, or the Direct Drive IGBT circuitry may short.

10. Monitor output at test points while a second person presses the Table Up pedal on the dock to raise the table. (Refer to the Caution at the beginning of this section for details.)

Note

This measurement is valid only during the period of table top upward motion. Once the table top reaches the maximum height, the dock motor will be too loaded to provide an accurate adjustment measurement.

11. Adjust R38 at the **RF/Pen Cabinet** and R80 at the **RF/Pen II Cabinet** and **RF/PDU Cabinet** to obtain a voltage waveform of six pulses per second $\pm 1/4$ pulse per second (Illustration 1-1 shows 5 and 3/4 pulses/second). Refer to Note above.

6 PULSES PER SECOND OSCILLOSCOPE DISPLAY

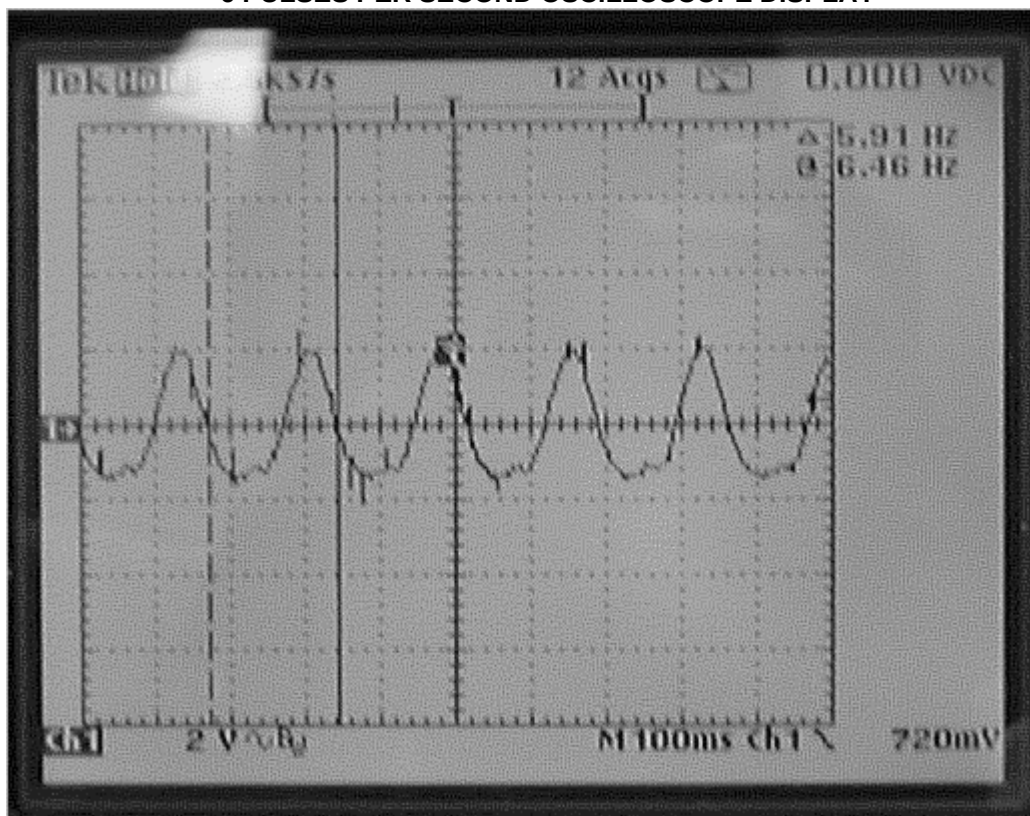


ILLUSTRATION 1-1

12. Repeat this procedure as necessary or whenever the Dock circuit board is replaced.

1-2 One-Man Procedure



The system must be set up for a Body mode protocol and the [Prepare to Scan] soft-key selected initially. Pulse Body mode once at TG of 0. Do NOT open the SSM top cover for one minute after the system has transitioned from Head, Surface, M/C or MNS mode to Body mode. Failure to follow these directions has resulted in the Dynamic Disable IGBT Circuitry (Direct Drive, Body1, Body2) overheating and shorting requiring CPD replacement.

1. Set up a Body mode protocol and select the [Prepare to Scan] softkey. Wait 5 minutes before opening the SSM cover. (Refer to above Caution for detail.)
2. Remove power to dock. Refer to the Safety procedure, *Section 6 - OSHA Lock-Out/Tag-Out Requirements* (SYSMSA2.DOC)
3. Lock-Out / Tag-Out circuit breaker MR1 A14 CB6 at rear of **RF/Pen Cabinet**, or circuit breaker MR1 A20 CB4 at rear of SSM in the **RF/Pen II** and **RF/PDU Cabinet**.
4. Verify that voltage between Dock and Dock RTN is **0 VDC** using a DVM:

RF/Pen Cabinet: Locate the MEPS Module at the bottom of the Cabinet, and the test point strip at the center of the rear panel of the MEPS module. Connect the DVM.

RF/Pen II and RF/PDU Cabinet: Locate the SSM at the bottom of the Cabinet and the test point strip at the center of the rear panel of the SSM. Connect the Digital Volt-Meter.

5. Remove the four screws and four No. 10 flat washers securing the MEPS or SSM module to the cabinet.
6. Slide the MEPS or SSM module out from the front of the cabinet.

RF/Pen Cabinet:

1. Loosen the three 4-40 captive screws on the top cover of the MEPS module. Open and remove the lid. Locate the High Current power supply board that contains the Voltage Potentiometer R38.
2. Pull out the RFSC and loosen the captive screws on the top cover of the SSM to gain access to the white plunger switch. Open the lid, and pull the High Voltage Enable Plunger Switch to the **Up** position.
3. Connect Channel 1 oscilloscope lead to TP3 (Dock 38.5V), and connect Channel 1 oscilloscope ground reference lead to TP1 or TP4 or TP7 (SGND).
4. Set Channel 1 trace to zero reference graticle on the oscilloscope display. Adjust time/div to approximately **100 mS/div**. Set oscilloscope channels to **AC Coupling**.

RF/Pen II Cabinet and RF/PDU Cabinet:

- a. Do **not** disable the HV at the front of the SSM.
- b. Remove the twelve 4-40 screws on the top cover of the SSM. Open the lid, and locate the Dock/Light power supply board that contains the Voltage Potentiometer R80.
- c. Connect Channel 1 oscilloscope lead to TP3 (Dock 38.5V), and connect Channel 2 oscilloscope lead to TP4 (DOCK-RTN).
- d. Connect both oscilloscope lead grounds together. Isolate ground leads with tape as required to ensure no contact with electrical circuitry.
- e. Set both channels to ground reference. Set Channel 2 to **add/invert**, and adjust both ground leads to the same zero reference graticle on the oscilloscope display. Adjust time/div to approximately **100 mS/div**. Set oscilloscope channels to **AC Coupling**.



SHOCK HAZARD! THE SSM IPS (INTEGRATED POWER SUPPLY) ASSEMBLY CIRCUITRY IS FLOATING. DO NOT REFERENCE TEST EQUIPMENT TO GROUND CHASSIS OR ANY GROUND TEST POINTS.

7. Lower Table to the full down position using the Table Down pedal.
8. Place a weight on the Table Up and Table Down foot pedals on the Dock. (The weighting of both pedals will ensure that the table does not top out and pull more current than normal.)
9. Switch on circuit breaker MR1 A14 CB6 in the **RF/Pen Cabinet** or on circuit breaker MR1 A20 CB4 in the **RF/Pen II** and **RF/PDU Cabinet**.

Note

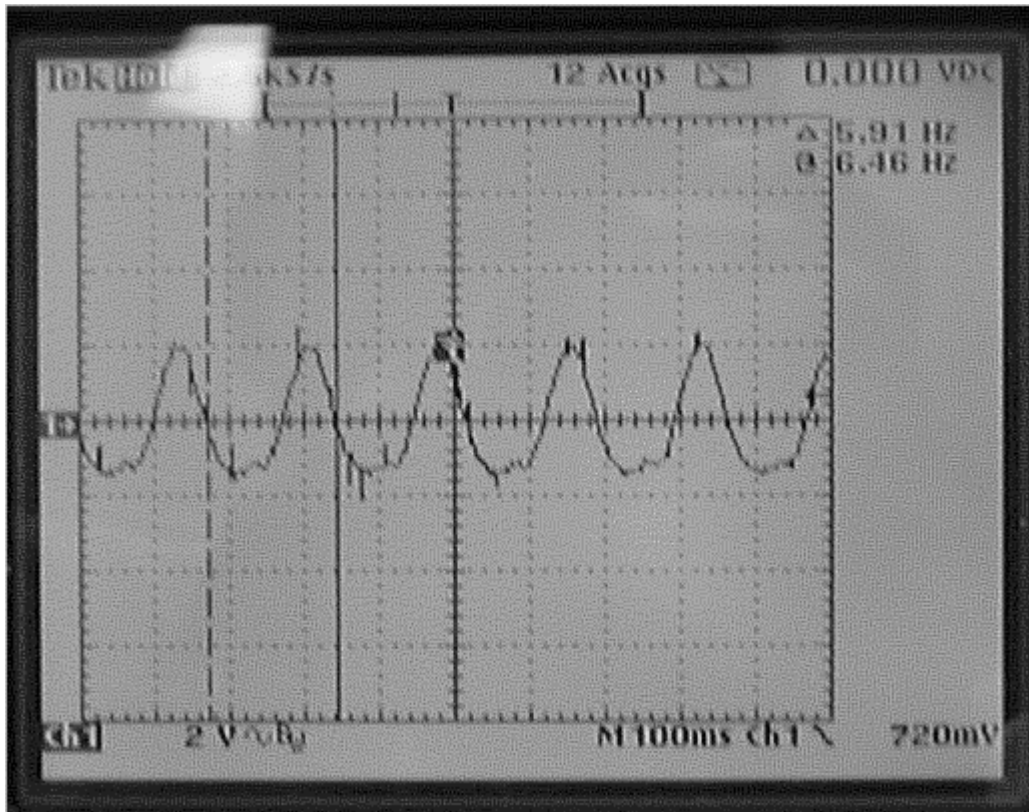
Only perform this procedure with the system set up for a Body mode protocol. (Refer to the Caution at the beginning of this section for details.)

10. Monitor output at test points (The following test setup measurement is valid only for approximately 12 seconds):

Note

This measurement is valid only during the period of table top upward motion. Once the table top reaches the maximum height, the dock motor will be too loaded to provide an accurate adjustment measurement.

11. Adjust R38 at the **RF/Pen Cabinet** and R80 at the **RF/Pen II Cabinet** and **RF/PDU Cabinet** to obtain a voltage waveform of six pulses per second $\pm 1/4$ pulse per second. (Illustration 1-2 shows 5 and 3/4 pulses/second.) Refer to Note above.



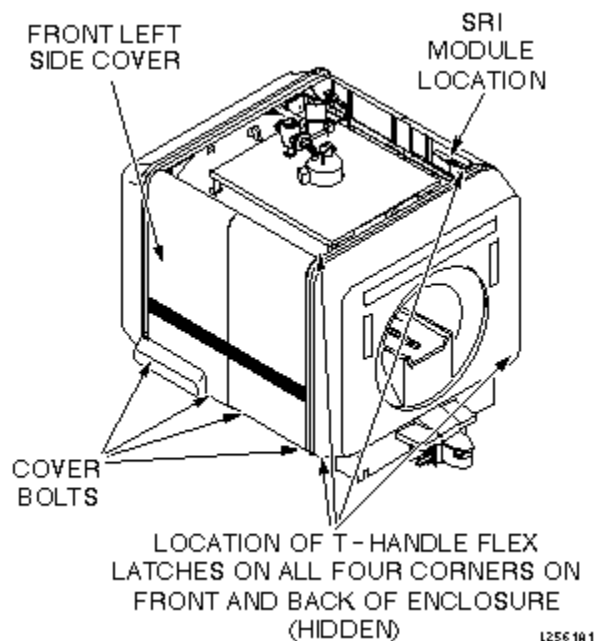
6 PULSES PER SECOND OSCILLOSCOPE DISPLAY
ILLUSTRATION 1-2

12. Repeat this procedure as necessary or whenever the Dock circuit board is replaced.

2- MAGNET ENCLOSURES

2-1 Non-CX Magnet Enclosure (MG2)

The magnet enclosure consists of six panels: front and back covers, front and back right side covers, and front and back left side covers. The side covers are removable. The front and rear covers pull out on slide rails, and can be removed only by disassembly. (See Illustration 2-1.)



ACCESS TO MAGNET ENCLOSURE
ILLUSTRATION 2-1

To gain access to all components inside the enclosure:

1. Disengage the flex latches on the top and bottom on each side of the front or back cover panel as desired. Slide out the cover panel to gain access to enclosure interior.
2. Remove the two screws from underside of side cover to be removed, swing bottom out, and lift covers off. Set covers aside.

2-2 Cx Magnet Enclosure (MG2)

The magnet enclosure consists of six panels: front and back covers, front and back right-side covers, and front and back left side covers. The side covers are removable. The front and rear covers pull out on slide rails, and can be removed only by disassembly.

Adjusting Enclosure Covers

There are jack screws attached to the front and back covers at the ends of the sliding rails. If these covers do not satisfactorily meet the front edges of front side covers or back edges of back side covers when sliding back into place, adjust the height using jack screws.

3- MOTOR ON SWITCH ACTUATION ADJUSTMENT

3-1 Description

The Motor On Actuation Switch in the dock assembly should be adjusted so it is activated when the Table Up foot pedal is pressed approximately 3/4 of its total length of travel. This is factory adjusted. However, it is possible to adjust the position of the Motor On switch in the field to maintain the proper actuation level.



FERROUS MATERIAL HAZARD! THE DOCK ASSEMBLY MAIN DOCK COVER CONTAINS A MOTOR MADE WITH FERROUS COMPONENTS. HOLDING THIS ASSEMBLY TOO CLOSE TO THE MAGNET BORE WILL CAUSE IT TO BE FORCIBLY ATTRACTED TO THE MAGNET. TO PREVENT POSSIBLE BODILY INJURY OR DAMAGE TO MOTOR, MAGNET ENCLOSURE OR MAGNET, PLACE DOCK ASSEMBLY ON FLOOR AND SLIDE IT OUT OF EXAM ROOM.

3-2 Procedure

1. Remove power to dock. Refer to the Safety procedure, *Section 6 - OSHA Lock-Out/Tag-Out Requirements (SYSMSA2.DOC)*.

A warning symbol consisting of the word "WARNING!" in bold, uppercase letters, enclosed within a rectangular border with a 3D effect.

WARNING!

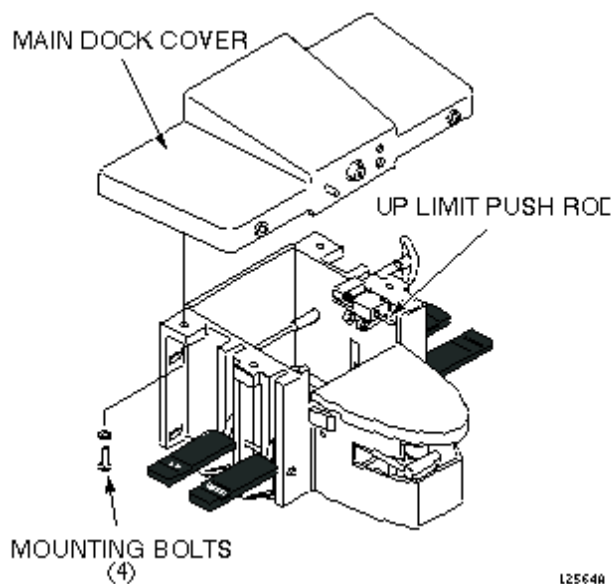
STRONG MAGNETIC FIELD! WHEN SERVICING ANY MAGNETIC EQUIPMENT, IT IS CRITICALLY IMPORTANT THAT THE SERVICE ENGINEER CONSCIOUSLY PLAN THE PATH TO BE TAKEN WHEN MOVING HIGHLY FERROUS DEVICES IN THE MAGNET ENVIRONMENT. THE PATH SHOULD BE AS FAR FROM THE MAGNET AS PRACTICAL.

SAFETY REQUIREMENTS:

- **THE SERVICE PATH MUST ALLOW FOR THE DOCK TO BE MOVED ALONG THE GROUND A DISTANCE EQUAL TO THAT OF THE DISTANCE BETWEEN THE MAGNET AND THE FOOT-END OF THE PATIENT TABLE PRIOR TO BEING LIFTED VERTICALLY.**
- **TWO (2) MR SAFETY TRAINED PERSONNEL MUST BE PRESENT AT ALL TIMES WHEN SERVICING HIGHLY FERROUS DEVICES IN THE AREAS OF MAGNETIC FIELDS.**

WHEN PLANNING A SERVICE PATH, IT IS CRITICAL THAT THE PATH BE CLEAR AND SUFFICIENTLY WIDE. ENSURE THAT THERE ARE NO TRIP HAZARDS, OBSTACLES, CLUTTER, SLIPPERY SURFACES OR OTHER ITEMS EVEN PARTIALLY RESTRICTING THE PATH. IF THERE ARE PORTABLE OBSTACLES IN A PATH, REMOVE THEM FROM THE AREA AND REPLACE THEM AFTER THE SERVICE ACTION IS COMPLETED. IT IS REQUIRED TO WALK THE PATH PRIOR TO BEGINNING SERVICE TO ENSURE THAT THERE IS SUFFICIENT SPACE THROUGH WHICH TO PASS FOR YOURSELF AND THE OBJECT BEING SERVICED.

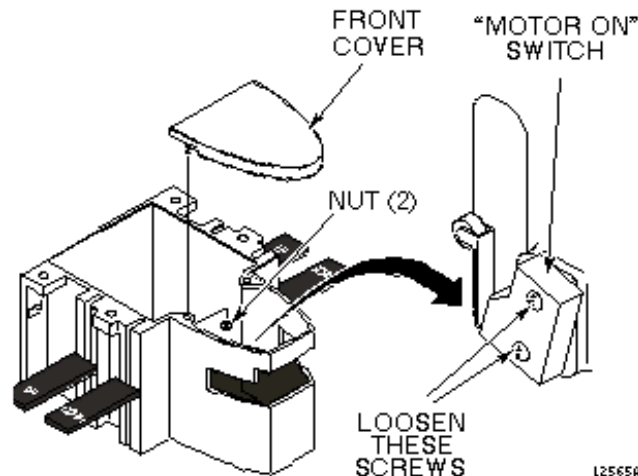
2. Remove the dock from the Magnet Room following instructions in *Section 3-1-1 in Dock Assembly Removal in SRI and Dock Hardware (ME1REA1.PDF)* for Dock Removal Instructions.
3. Once dock is removed from the Magnet Room, remove the four mounting bolts from the main dock cover shown in Illustration 3-1.



DOCK COVER REMOVAL
ILLUSTRATION 3-1

4. Slide main dock cover forward until Up Limit Switch pushrod just clears the main dock cover holes (Illustration 3-1).

5. Remove front cover by removing two nuts (shown in Illustration 3-2) securing front cover to the dock.



MOTOR ON SWITCH ADJUSTMENT
ILLUSTRATION 3-2

6. Loosen two mounting screws securing Motor On Switch to side of dock assembly. (See Illustration 3-2.)
7. Adjust Motor On Switch position so it is activated (clicks) when the Table Up foot pedal is pressed approximately 3/4 of its total length of travel.
8. After adjustment is complete, tighten the two mounting screws.
9. Reinstall front cover by attaching the two mounting nuts.
10. Carefully position main dock cover so that Up Limit Switch pushrod is aligned with upper hole in main dock cover.
11. Replace main dock cover by securing four mounting bolts (Illustration 3-1).



FERROUS MATERIAL HAZARD! THE DOCK ASSEMBLY MAIN DOCK COVER CONTAINS A MOTOR MADE WITH FERROUS COMPONENTS. HOLDING THIS ASSEMBLY TOO CLOSE TO THE MAGNET BORE WILL CAUSE IT TO BE FORCIBLY ATTRACTED TO THE MAGNET. TO PREVENT POSSIBLE BODILY INJURY OR DAMAGE TO MOTOR, MAGNET ENCLOSURE, OR MAGNET, PLACE DOCK ASSEMBLY ON FLOOR AND SLIDE IT OUT OF EXAM ROOM.

12. Reinstall the dock in the Magnet Room following *Section 3-1-2 Dock Assembly Installation* in *SRI and Dock Hardware* (ME1REA1.PDF)
13. Ensure that dock Motor On Switch operates properly when Table Up pedal is pressed.

4- DOCK MOTOR GEAR ASSEMBLY ADJUSTMENT

The dock motor gear box has a clutch that prevents backlash on the gears in the gear box. This clutch is not intended to regulate dock motor speed. (Refer to *Section 1- Dock Motor Speed Adjustment*.) This clutch simply prevents gear backlash that would make the gear assembly noisy.

There are three probable causes of a noisy gear assembly:

1. Cam improperly adjusted
2. Excessive play on motor tube spring
3. Clutch improperly adjusted

Perform procedures in Sections 4-1, 4-2 and 4-3 in the sequence presented, to correct excessive gear assembly noise.

4-1 Cam Adjustment

1. Remove power to the dock. Refer to the Safety procedure, *Section 6 - OSHA Lock-Out/Tag-Out Requirements* (SYSMSA2.DOC).

WARNING!

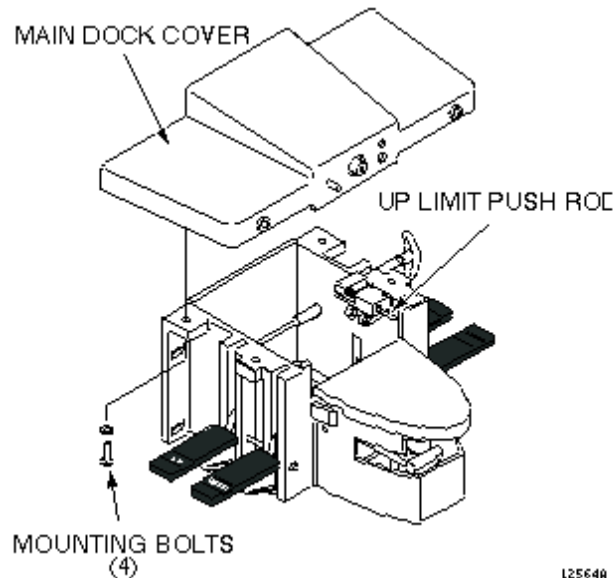
STRONG MAGNETIC FIELD! WHEN SERVICING ANY MAGNETIC EQUIPMENT, IT IS CRITICALLY IMPORTANT THAT THE SERVICE ENGINEER CONSCIOUSLY PLAN THE PATH TO BE TAKEN WHEN MOVING HIGHLY FERROUS DEVICES IN THE MAGNET ENVIRONMENT. THE PATH SHOULD BE AS FAR FROM THE MAGNET AS PRACTICAL.

SAFETY REQUIREMENTS:

- **THE SERVICE PATH MUST ALLOW FOR THE DOCK TO BE MOVED ALONG THE GROUND A DISTANCE EQUAL TO THAT OF THE DISTANCE BETWEEN THE MAGNET AND THE FOOT-END OF THE PATIENT TABLE PRIOR TO BEING LIFTED VERTICALLY.**
- **TWO (2) MR SAFETY TRAINED PERSONNEL MUST BE PRESENT AT ALL TIMES WHEN SERVICING HIGHLY FERROUS DEVICES IN THE AREAS OF MAGNETIC FIELDS.**

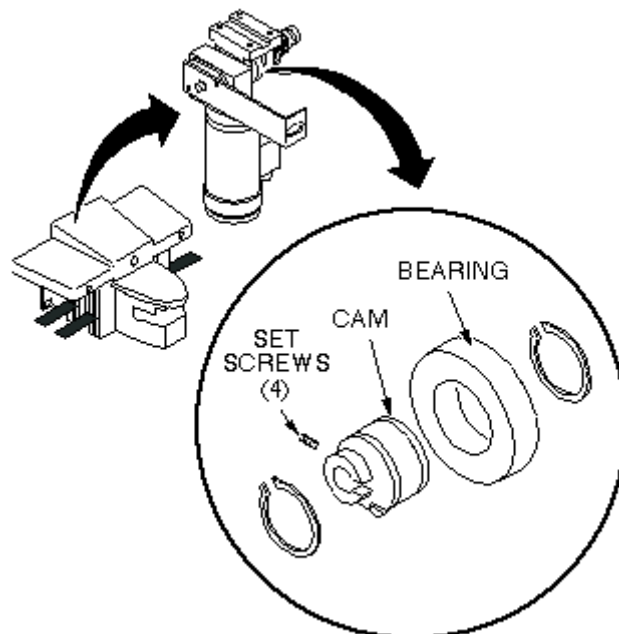
WHEN PLANNING A SERVICE PATH, IT IS CRITICAL THAT THE PATH BE CLEAR AND SUFFICIENTLY WIDE. ENSURE THAT THERE ARE NO TRIP HAZARDS, OBSTACLES, CLUTTER, SLIPPERY SURFACES OR OTHER ITEMS EVEN PARTIALLY RESTRICTING THE PATH. IF THERE ARE PORTABLE OBSTACLES IN A PATH, REMOVE THEM FROM THE AREA AND REPLACE THEM AFTER THE SERVICE ACTION IS COMPLETED. IT IS REQUIRED TO WALK THE PATH PRIOR TO BEGINNING SERVICE TO ENSURE THAT THERE IS SUFFICIENT SPACE THROUGH WHICH TO PASS FOR YOURSELF AND THE OBJECT BEING SERVICED.

2. Remove dock from the Magnet Room following instructions in *Section 3-1-1 Dock Assembly Removal in SRI and Dock Hardware (ME1REA1.PDF)*.
3. Remove the four mounting bolts from main dock cover. (See Illustration 4-1.)



DOCK COVER REMOVAL
ILLUSTRATION 4-1

4. Slide the main dock cover forward until Up Limit Switch pushrod just clears dock cover holes (Illustration 4-1).
5. Check the four set screws (shown in Illustration 4-2) on the motor cam. If the set screws are loose, apply Loctite® 242 and Primer T to set screws and tighten. Allow 15 minutes to harden.



DOCK MOTOR CAM ADJUSTMENT
ILLUSTRATION 4-2

6. Carefully position main dock cover so that Up Limit switch push rod is aligned with upper hole in dock cover (Illustration 4-1).
7. Replace the main dock cover and secure with four mounting bolts shown in Illustration 4-1.

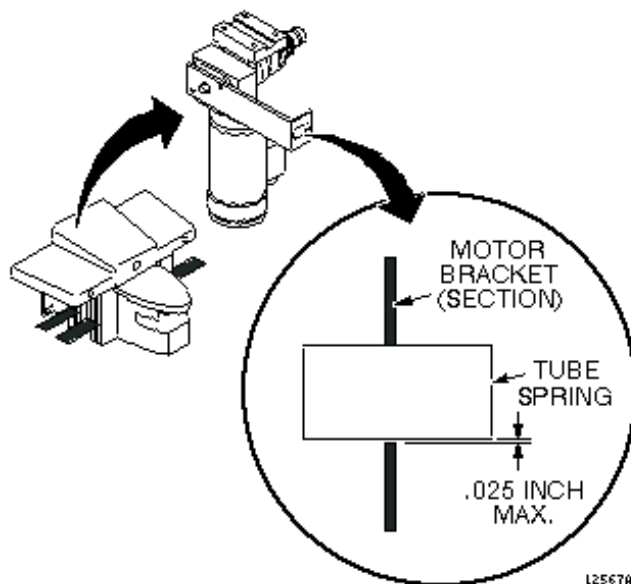


FERROUS MATERIAL HAZARD!! THE DOCK ASSEMBLY MAIN DOCK COVER CONTAINS A MOTOR MADE WITH FERROUS COMPONENTS. HOLDING THIS ASSEMBLY TOO CLOSE TO THE MAGNET BORE WILL CAUSE IT TO BE FORCIBLY ATTRACTED TO THE MAGNET. TO PREVENT POSSIBLE BODILY INJURY OR DAMAGE TO MOTOR, MAGNET ENCLOSURE, OR MAGNET, PLACE DOCK ASSEMBLY ON FLOOR AND SLIDE IT OUT OF EXAM ROOM.

8. Reinstall the dock in the Magnet Room following instructions in *Section 3-1-2 Dock Assembly Installation* in *SRI and Dock Hardware* (ME1REA1.PDF).
9. Exercise the Table Up pedal to see if gear noise is eliminated. If not, remove dock assembly as detailed in Section 4-1 and proceed *Section 4-2 Motor Tube Spring Check*.

4-2 Motor Tube Spring Check

1. Remove the dock from the Magnet Room following instructions in *Section 3-1-1 Dock Assembly Removal* in *SRI and Dock Hardware* (ME1REA1.PDF).
2. Check the play between tube spring and motor mounting bracket. (See Illustration 4-3.)



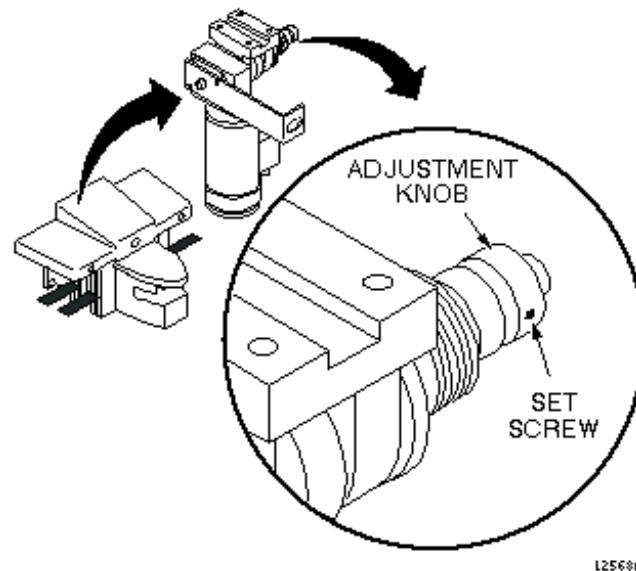
**MOTOR TUBE SPRING
ILLUSTRATION 4-3**

3. If play is more than 0.025 inch (0.6 mm), replace tube spring.

4. Replace the main dock cover on the dock assembly, move into Magnet Room, heeding danger precautions on previous page.
5. Reinstall the dock in the Magnet Room following instructions in *Section 3-1-2 Dock Assembly Installation* in *SRI and Dock Hardware* (ME1REA1.PDF).
6. Exercise the Table Up pedal to ensure gear noise is eliminated. If not, remove dock assembly as detailed in section 4-1 and perform section 4-3 Clutch Adjustment.

4-3 Clutch Adjustment

1. Remove the dock from the Magnet Room following instructions in *Section 3-1-1 Dock Assembly Removal* in *SRI and Dock Hardware* (ME1REA1.PDF).
2. Remove main dock cover outside of the scan room as outlined in Section 4-1.
3. Loosen the set screw on the clutch adjustment knob (Illustration 4-4).



DOCK MOTOR CLUTCH ADJUSTMENT
ILLUSTRATION 4-4

4. Adjust knob two clicks tighter (clockwise), and tighten the set screw.

Note

Do **not** adjust clutch more than two clicks at a time. The clutch should not be tightened to the point where it affects the dock motor function.

5. Replace the main dock cover on the dock assembly as described in Section 4-2.
6. Reinstall the dock in the Magnet Room following instructions in *Section 3-1-2 Dock Assembly Installation* in *SRI and Dock Hardware* (ME1REA1.PDF).
7. Exercise the Table Up pedal to ensure gear noise is eliminated. If not, remove dock assembly and repeat adjustment procedures as detailed in Sections 4-1 through 4-3.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	June 16, 1998	J. Saperstein	Initial conversion from Toolbook to Word.
1	January 6, 1999	K. Keshena	Added info for location of breaker in RF/PDU cabinet. Updates to styles.
2	May 21, 1999	SM Atladottir	Updated procedure references for new GUI
3	September 2, 1999	Resa Lambert	Moved Dock Speed Adjustment to section1, corrected procedure and added Illustration. Added SSM Body mode protocol Caution.
4	September 2010	B. Kohl	Changes made per CAPA 1859392. Section 3-1 Procedure, updated removal and installation sections. Section 4-1 Cam Adjustment Procedure, updated removal and installation sections. Section 4-2 Motor Tube Spring Check, updated procedure. Section 4-3 Clutch Adjustment, updated procedure.